

1. EXECUTIVE OVERVIEW & SYSTEM ARCHITECTURE

The **MSU Internal Audit Management System** is an enterprise governance, risk, and audit execution platform tailored for Midlands State University. Built to uphold the standards of the Institute of Internal Auditors (IIA), the system digitalizes the end-to-end audit lifecycle from strategic annual planning down to finding remediation.

Core Modules & Capabilities:

- **Audit Universe:** Annual rolling audit planning, engagement scheduling, and multi-auditor team allocation.
- **Risk Management:** Integrated Risk Register, Inherent & Residual risk scoring, and interactive 5x5 Heatmap.
- **Working Papers & Fieldwork:** Electronic audit testing papers, methodology templates, and evidentiary document storage.
- **Finding Tracker:** Comprehensive deficiency cataloging, 5-level risk escalation, and root-cause mapping.
- **Action Items:** Management corrective action tracking, automated overdue alerts, and remediation follow-up.
- **Reports Centre:** 4-stage review and approval governance, transmittal tracking, and official PDF generation.
- **Executive Governance:** Role-tailored dashboards for the Vice Chancellor, Audit Committee, and University Council.

2. SIGN-OFF AUTHORITY & APPROVAL LEVELS (1-5)

The system enforces a dual-layer security model comprising **System Roles** (interface permissions) and **Approval Levels** (hierarchical sign-off authority and delegation limits):

LEVEL	POSITION / TIER	SIGN-OFF AUTHORITY & OPERATIONAL SCOPE
LEVEL 1	Field / Operational Auditor Junior Auditor, Audit Assistant	• Executes test procedures and records fieldwork data. • Prepares working papers and uploads supporting audit evidence. • Drafts initial audit observations, root causes, and draft reports. • <i>Does not have authority to approve plans or issue final reports.</i>
LEVEL 2	Supervisory Reviewer Senior Auditor, Risk Officer	• Performs first-line quality review on working papers and audit evidence. • Reviews draft audit reports prepared by Level 1 auditors. • Logs required corrections or endorses reports for Chief Auditor approval. • Validates enterprise risk ratings and escalates critical findings.
LEVEL 3	Management / Chief Auditor Audit Manager, Chief Internal Auditor	• Approves Audit Universe engagements and authorizes audit commencement. • Conducts executive reviews on draft reports and endorses quality compliance. • Formally authorizes and issues Official Final Audit Reports. • Verifies management remediation and formally closes resolved findings.
LEVEL 4	Executive & Governance Audit Committee, Vice Chancellor, Council	• High-level university governance oversight. • Accesses Executive & Governance Dashboards and Quarterly Meeting Packs. • Evaluates university-wide strategic risk posture, audit cycle times, and unresolved high-risk exposures.
LEVEL 5	System Administrator System Admin, IT Directorate	• Root configuration, staff account provisioning, and role assignment. • System integration management (Google SSO, ERP budget linkage, Risk sync). • Monitors immutable system audit logs and application health.

3. OPERATIONAL WORKFLOW GUIDE BY ROLE & LEVEL

Level 1: Operational Field Auditor Workflow

- 1. **Assignment:** Receives allocation to an active audit engagement in the *Audit Universe*.
- 2. **Fieldwork Testing:** Navigates to *Working Papers*, creates working papers associated with linked university risks, and attaches test evidence files.
- 3. **Deficiency Logging:** Identifies internal control gaps and creates records in the *Finding Tracker* (noting criteria, condition, cause, effect, and recommendation).
- 4. **Draft Report Preparation:** Upon conclusion of testing, opens *Reports Centre* → *Prepare New Report*, fills the executive summary, methodology, and transmittal comments.
- 5. **Draft Submission:** Submits the draft report to the Senior Internal Auditor (Status becomes `pending_senior_review`, and the **Draft Issue Date** is captured).
- 6. **Addressing Feedback:** If returned with comments, addresses the requested corrections and resubmits.

Level 2: Senior Internal Auditor (Supervisory Review) Workflow

- 1. **Quality Review:** Accesses assigned engagements and reviews submitted working papers and test sampling.
- 2. **Report Review Gate:** Opens the draft report in *Reports Centre*. Evaluates completeness and accuracy.
- 3. **Decision Pathways:**
 - *If corrections are required:* Clicks "**Request Corrections**", enters specific observations. The report status updates to `returned_for_revision`.
 - *If quality standards are met:* Clicks "**Endorse & Forward to Chief**" with senior endorsement notes. The report moves to `pending_chief_approval`.
- 4. **Finding Escalation:** Evaluates finding risk ratings and triggers formal escalation levels if management response is overdue.

Level 3: Audit Manager / Chief Internal Auditor Workflow

- 1. **Annual Planning:** Creates audit engagements in the *Audit Universe*, links ERP budget codes, and defines planned dates.
- 2. **Team Allocation:** Uses the "**Assign Audit Team**" tool to allocate Lead Auditors and team members to engagements.
- 3. **Engagement Authorization:** Reviews engagement charters and approves draft plans (Status: `planned` ? `in_progress`).
- 4. **Chief Report Review:** Reviews reports forwarded by Senior Auditors.
- 5. **Final Report Issuance:** Approves the report and clicks "**Authorize & Issue Final Report**" (Captures the **Final Issue Date**, locks the report as `final_issued`, marks the audit completed, and watermarks the official PDF).
- 6. **Finding Closure:** Reviews verified action items and formally marks findings as resolved/closed.

Level 4: Executive Management, Audit Committee & Council Workflow

- 1. **Executive Dashboard:** Reviews real-time KPIs covering university risk coverage %, audit cycle times, and finding resolution velocity.
- 2. **Quarterly Meeting Pack:** Downloads pre-compiled executive meeting packs containing summarized finding aging and high-risk exposures.
- 3. **Risk Heatmap Analysis:** Analyzes dynamic risk distribution across faculties, departments, and compliance mandates.

Level 5: System Administrator & IT Lead Workflow

- 1. **Staff Account Onboarding:** Creates user profiles with mandatory `@staff.msu.ac.zw` emails, positions, and approval levels (1-5).
- 2. **Google SSO Governance:** Enforces secure passwordless Google Workspace authentication.
- 3. **Audit Trail Inspection:** Reviews immutable system logs in *System Logs* to trace every login, record creation, update, and deletion.

4. COMPREHENSIVE AUDIT ENGAGEMENT LIFECYCLE

STAGE	KEY ACTIVITIES	PRIMARY ACTORS	SYSTEM OUTPUT / DELIVERABLE
1. Planning	Engagement creation, risk linkage, ERP budget coding	Audit Manager (L3), System Admin (L5)	Approved Audit Universe Plan (Code: AUD-YYYY-XXXX)
2. Allocation	Assigning auditors & field team members	Chief Auditor / Audit Manager (L3)	Engagement Team roster with avatar display
3. Fieldwork	Executing tests, collecting evidence, sampling	Auditors / Field Team (L1)	Versioned Working Papers & uploaded evidence
4. Findings	Identifying deficiencies & root causes	Auditors (L1), Senior Reviewer (L2)	Finding records with severity & action items
5. Draft Report	Drafting summary, methodology & initial issue	Audit Team Preparer (L1)	Draft Report (Captures Draft Issue Date)
6. Senior Review	Quality testing, correction logs, endorsement	Senior Internal Auditor (L2)	Endorsed Report or Correction Tracker items
7. Chief Approval	Executive review, authorization, final issue	Chief Internal Auditor (L3)	Official Final Report (Final Issue Date)
8. Remediation	Action plan follow-up, evidence review, closure	Action Owners, Audit Manager (L3)	Closed findings & Governance Analytics

5. AUTHENTICATION & SECURITY POLICY

- 1. Single Sign-On (Google Workspace):** All university staff authenticate securely via Google OAuth2 with their official `@staff.msu.ac.zw` credentials.

2. Granular Role-Based Access Control (RBAC): Managed through Spatie permissions to guarantee separation of duties.

3. Tamper-Proof Audit Trails: Every critical model action (create, update, delete, approve, issue) is logged with the user ID, timestamp, and before/after state diff.